A Paleomagnetic Investigation of Some of the Precambrian Igneous Rocks of Southeast Missouri

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Abstract. Measurements of the magnetic susceptibility and the intensity and direction of remanent magnetization were made on 325 2.5-cm cylindrical cores prepared from 189 samples of the felsites, granites, and diabases exposed in the Ironton, Fredericktown, and Coldwater quadrangles, Missouri. These measurements show, first, that the directions of remanent magnetization, although quite variable, are significantly different from the direction of the present geomagnetic field in southeast Missouri and, second, that partial demagnetization in an ac field of 300 to 700 oe greatly improves the initial scatter of the remanent magnetization vector by diminishing the anomalous components of viscous and isothermal remanent magnetization shown by many samples. The paleomagnetic data from five rock units indicate a Precambrian geomagnetic pole in the vicinity of 150°W and 5°N.

Introduction. The St. Francois Mountains of southeast Missouri encompass about 2500 mi² in Iron, Madison, and St. Francois Counties (see Figure 1). These mountains, which are characterized by rolling, knob-like hills of Precambrian igneous rocks, stand some 500 to 1700 feet in relief above the surrounding terrain.

The igneous rocks of this area have been extensively investigated petrologically by *Tolman and Robertson* [1960] and others. The present work represents the first extensive effort to investigate the paleomagnetic properties of these relatively undisturbed Precambrian rocks.

Geology and sampling. The Precambrian igneous rocks of the St. Francois Mountains fall into three major petrologic groups: a variegated series of granites, a system of basic intrusives of gabbroic composition, and a group of intrusive and extrusive felsites of rhyolitic composition. The structural relationships between these major petrologic units are complex. On the basis of extensive field work by *Tolman and Robertson* [1960] and others, the general geologic sequence was the initial extrusion of a thick series of felsitic volcanic flows. The flows were intruded by a granite batholith with its associated masses of intrusive felsites. Basic dikes and sills and irregular stocks and bosses subsequently intruded into both the granites and felsites.

The approximate Rb/Sr age of the granites has been reported to be 1300 ± 100 m.y. by *Hayes* [1961] and determined to range from 1300 to 1430 m.y. by *Tilton et al.* [1962].

One hundred and eigthy-nine samples of the various felsite, granite, and diabase rock units (see Table 1) were collected from suitable exposures in a three-quadrangle area. Each sample was oriented in situ with respect to geographic north and the horizontal plane. Orientation lines were scribed on each sample, and the sample was chiseled from the outcrop or exposure. In the laboratory, one or more cylindrical cores, 2.5 cm in length and 2.5 cm in diameter, were prepared from each sample using a diamond-impregnated core barrel and a small Dimet saw.

Paleomagnetic results. The basic measurements made on each sample were: (1) the direction and intensity of remanent magnetization; (2) the magnitude of magnetic susceptibility; and (3) the demagnetization characteristics. All measurements reported were made either in Silver Spring, Maryland, or Menlo Park, California, at the Geophysics Laboratory of the U. S. Geological Survey. The direction and intensity of magnetization were measured with a spinner-type magnetometer patterned after the magnetometer developed

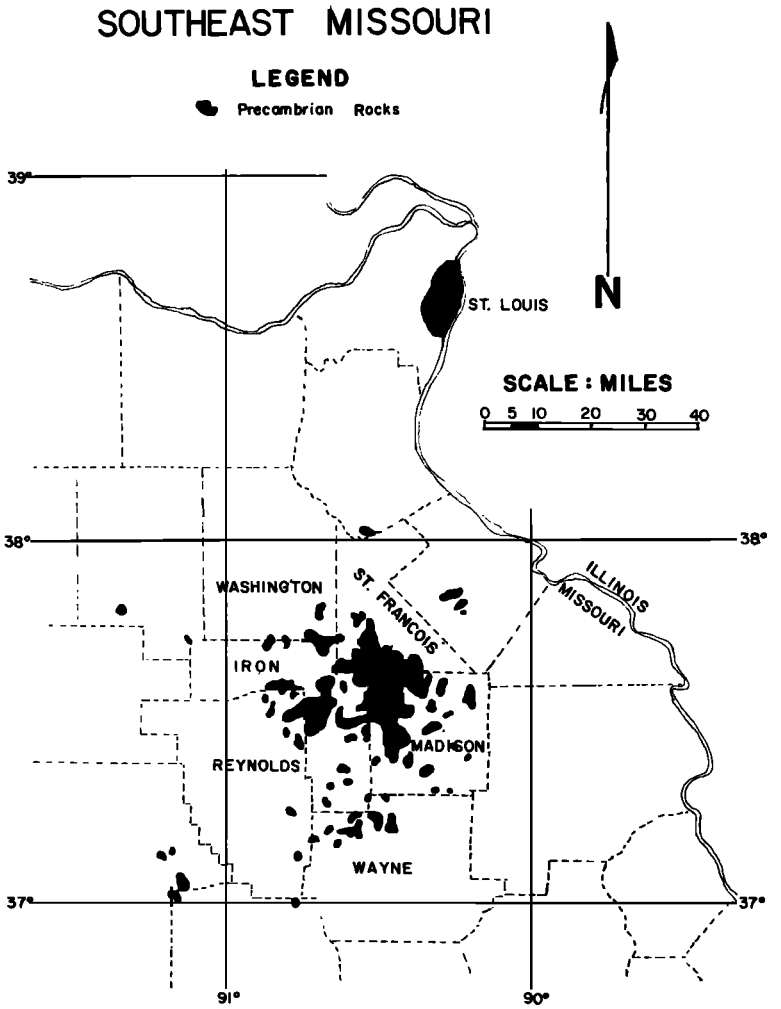


Fig. 1. Index map showing location of Precambrian igneous exposures in southeast Missouri.

by *Johnson* [1938]. The susceptibility was measured with a Mooney portable susceptibility meter. The samples were partially demagnetized in an ac magnetic field (see *Cox and Doell* [1960, p. 659] for a description of the process). The results of these measurements are given in Table 1.

Samples of the Stouts Creek rhyolite were collected from scattered outcrops over a 15 mi^2 area. The directions of remanent magnetization (Figure 2) are greatly scattered and exhibit both normal and reversed polarity. For most samples, both the intensity and direction of magnetization changed significantly after partial demagnetization in an ac field. In most cases, a peak field of 200 to 300 oe was sufficient to remove the soft components of magnetization residing in low coercive force domains. Continued partial demagnetization in greater fields failed to change significantly either the direction or intensity of magnetization remaining in these samples—an indication that the magnetization component remaining in the sample resides in high coercive force domains. In some samples, however, the intensity and direction of magnetization changed continuously throughout a partial demagnetization cycle of 0 to 700 oe—an indication that the magnetization components are equally distributed throughout the entire coercive force domain. Two of the most typical de-

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TABLE 1. Average Magnetic Properties

Rock	N	k ($\times 10^{-3}$)	M_n J_0 ($\times 10^{-3}$)	M_n D, deg	M_n I, deg	M_n α_{95}, deg	M_{ac} N	M_{ac} D, deg	M_{ac} I, deg	M_{ac} α_{95}, deg	Alternating Field, oe
Stouts Creek rhyolite	103	0.43	4.49	244.8	61.2	19.5	46	243.3	49.3	8.9	300–700
Royal Gorge rhyolite	16	0.25	0.62					256.8	37.6	51.5	450
Hogan Mountain rhyolite	14	0.14	0.57	290.7	31.8	35.8	11	243.7	42.5	14.8	450
Skrainka diabase	12	2.04	2.69	246.7	42.3	11.0	12	235.5	39.6	8.5	300
Granites	34	0.79	No values obtained								
Iron Mountain andesite	10	0.20	20.00	244.5	58.0	17.1					0–400

Statistical techniques after *Fisher* [1953].
N = number of samples measured.
k = magnetic susceptibility.
J_0 = intensity of remanent magnetization (at 0 oe).
D = declination of mean direction of remanent magnetization.
I = inclination of mean direction of remanent magnetization.
α_{95} = semiangle of cone of 95% confidence for mean direction.
M_n = denotes natural remanent magnetization.
M_{ac} = denotes magnetization after treatment in ac field.

magnetization curves measured in Stouts Creek samples are shown in Figure 2. Their most prominent feature is the rapid decrease of intensity of magnetization with increase in demagnetization field strength. The direction of magnetization changed significantly in both samples up to the field strength where the intensity tends to remain constant. Almost without exception the samples which displayed this type of partial demagnetization were collected at or near the top of a hill, usually from a well-exposed outcrop. The rapid decrease in intensity,

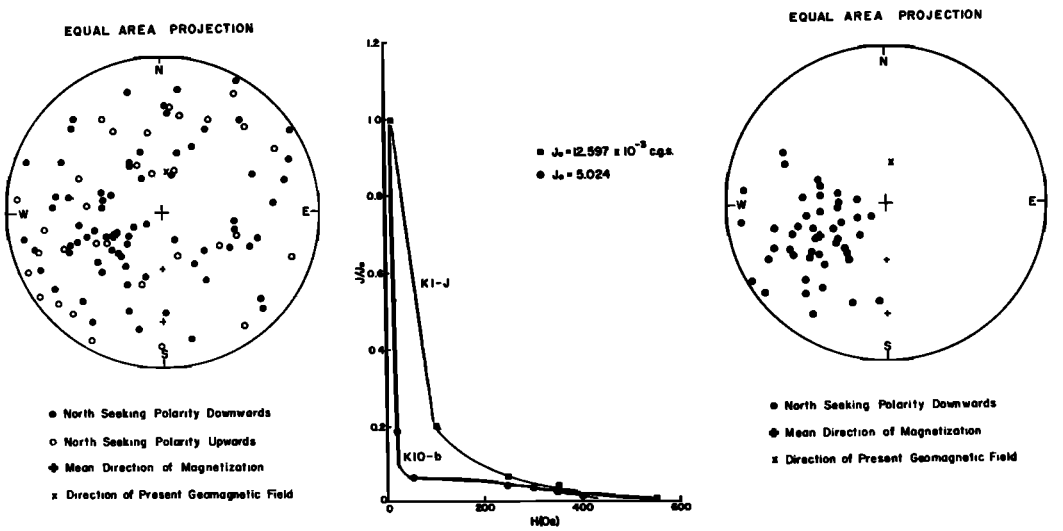


Fig. 2. (Left) Directions of remanent magnetization measured in Stouts Creek rhyolite samples, (center) two representative curves of partial demagnetization for Stouts Creek rhyolite, and (right) directions of remanent magnetization measured in Stouts Creek rhyolite samples after partial demagnetization (samples which show instability are not included).

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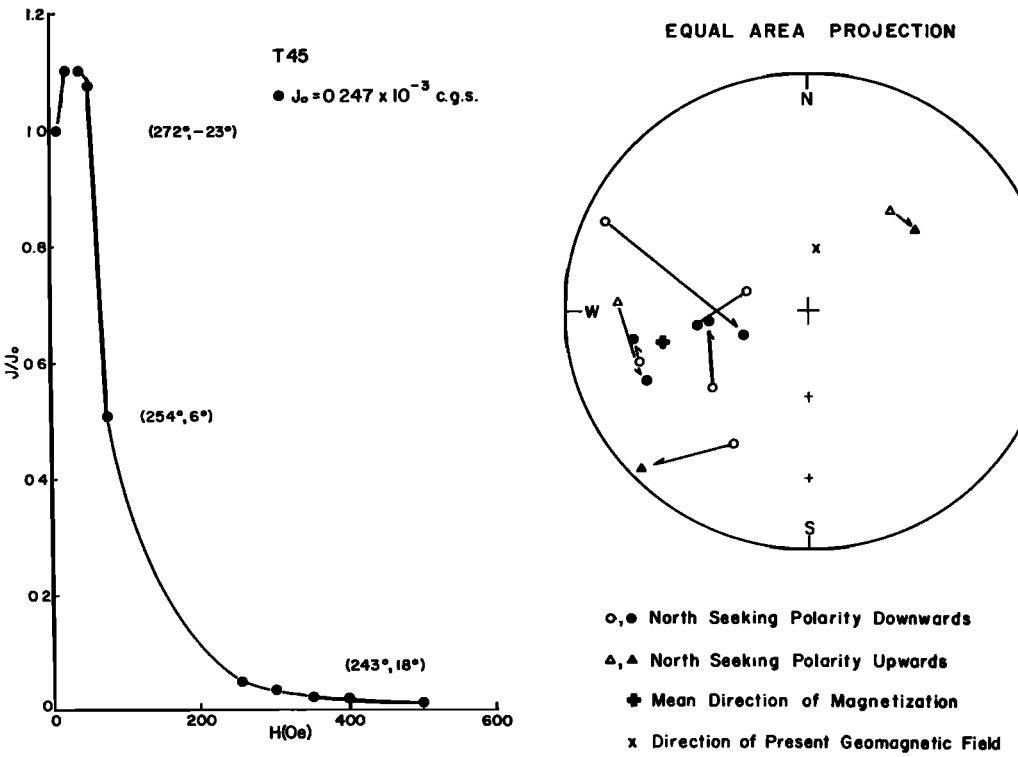


Fig. 3. (Left) A representative partial demagnetization curve of the Royal Gorge rhyolite; (right) directions of remanent magnetization before partial demagnetization (open figures) and after partial demagnetization (solid figures).

usually with a significant change in direction of magnetization, suggests strongly that lightning has imparted an anomalous component of IRM on the natural TRM component and modified both the intensity and direction of magnetization in the sample. Fortunately, the IRM component resides primarily in the low coercive force domains in most samples, and its effects can be minimized by partial demagnetization.

Samples of a second felsite rock unit, the Royal Gorge rhyolite, were collected over an area of approximately 5 mi^2. As with the Stouts Creek samples, the directions of magnetization are randomly scattered (see Figure 3). The partial demagnetization experiments indicate an anomalous component of viscous magnetization superposed on the TRM component in many cases. The demagnetization curve (Figure 3) is a typical example of this phenomenon, i.e., a characteristic initial rise and then a sharp drop in intensity as the peak demagnetization field is increased. The direction of magnetization changes significantly during the demagnetization cycle; being 'reversed' initially, it gradually became normal at a peak field of approximately 75 oe and then changed directions only slightly as the peak field was increased from 75 to 400 oe.

Fourteen samples of a third felsite unit, the Hogan Mountain rhyolite, were collected from outcrops over an area of approximately 8 mi^2. The directions of remanent magnetization for each sample before and after partial demagnetization are plotted in Figure 4. The mean direction before demagnetization is significantly different from the direction of the present geomagnetic field, in spite of the large scatter, but is greatly improved after demagnetization.

Twelve samples of a shallow diabase still (Skrainka diabase) were collected from various points in an abandoned quarry, and the results of the measurements are indicated in Figure 5. Measurements of the initial NRM show the directions of magnetization to be closely grouped; however, after partial demagnetiza-

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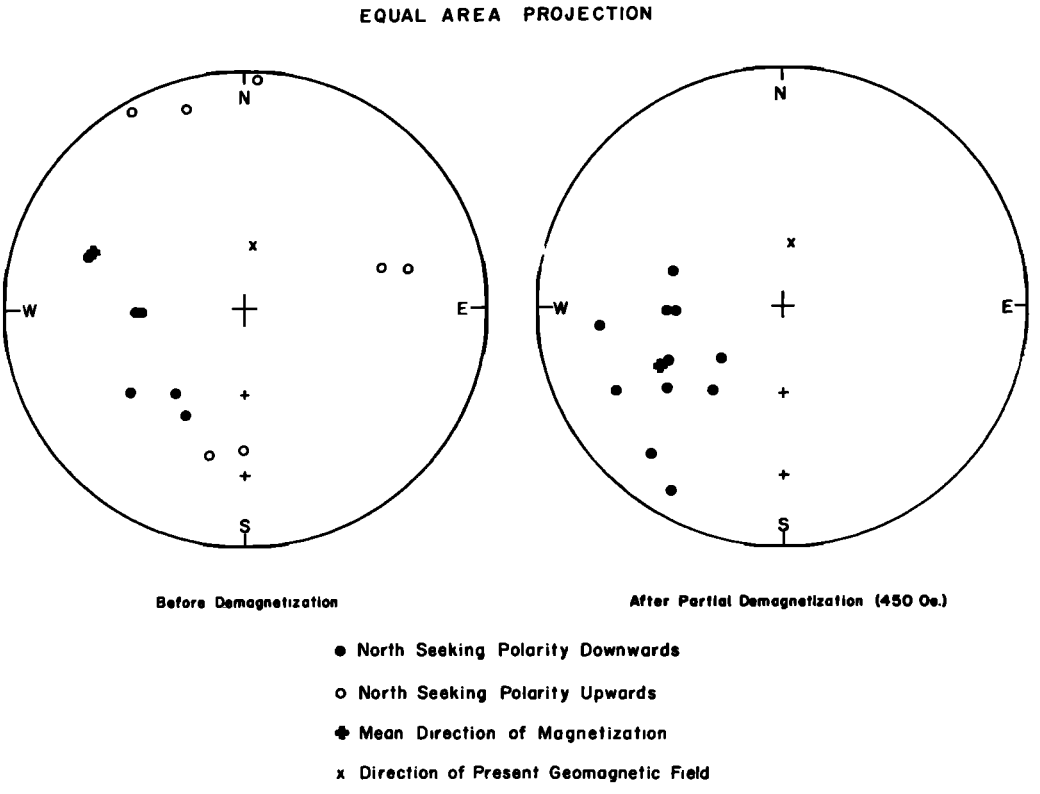


Fig. 4. (Left) Directions of remanent magnetization of the Hogan Mountain rhyolite before partial demagnetization; (right) directions of remanent magnetization of Hogan Mountain rhyolite after partial demagnetization.

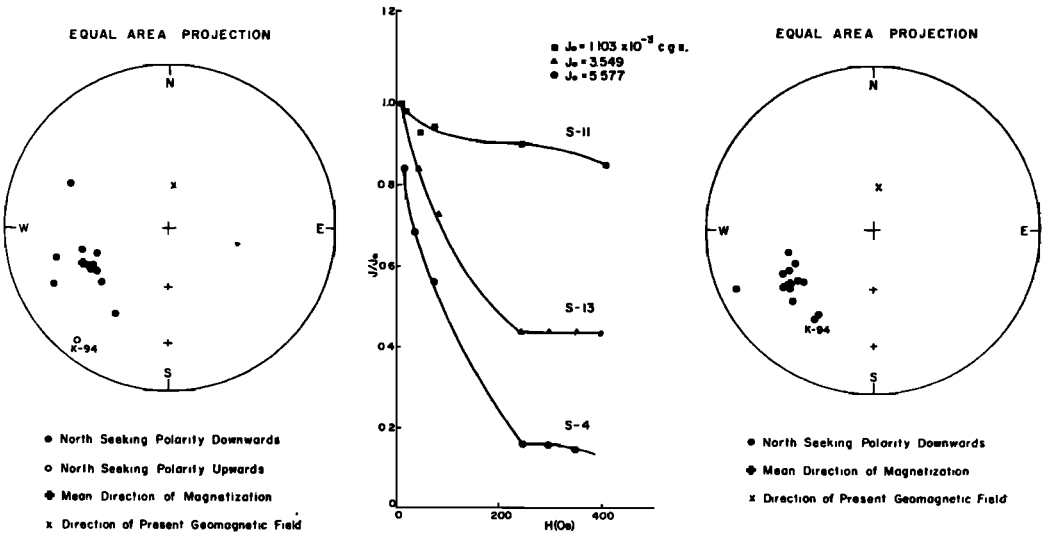


Fig. 5. (Left) Directions of remanent magnetization of Skrainka diabase before partial demagnetization, (center) typical partial demagnetization curves for the Skrainka diabase, and (right) directions of remanent magnetization after partial demagnetization.

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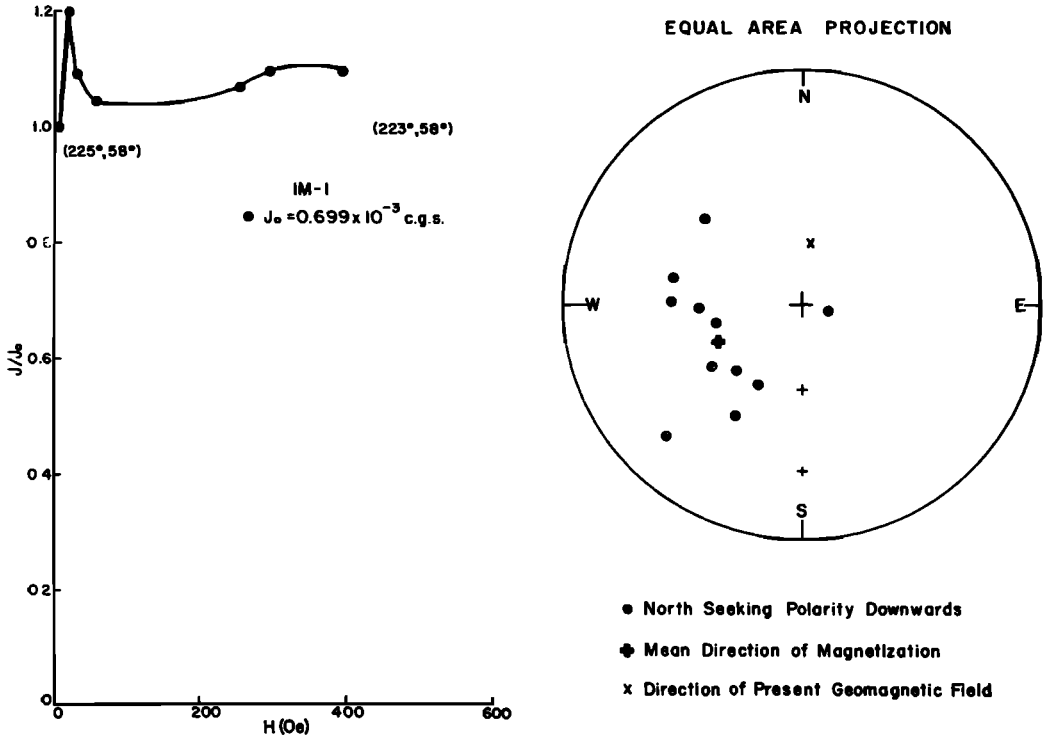


Fig. 6. (Left) A representative partial demagnetization curve for the Iron Mountain andesite; (right) directions of remanent magnetization measured in the Iron Mountain andesite samples.

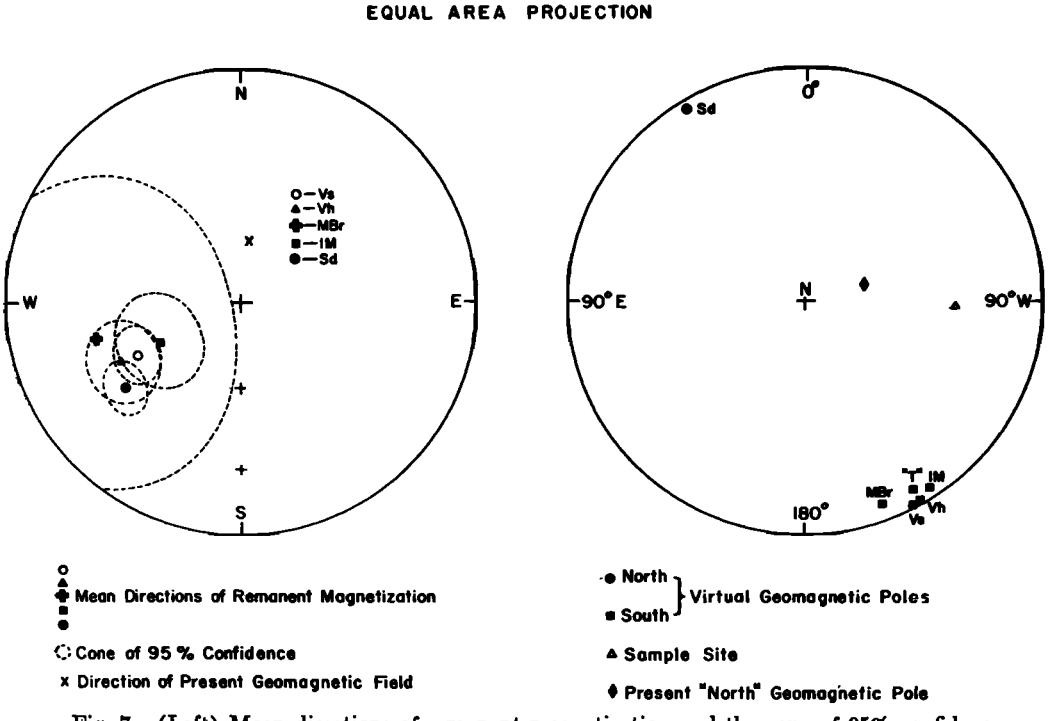


Fig. 7. (Left) Mean directions of remanent magnetization and the cone of 95% confidence for each Precambrian rock unit sampled; (right) the mean virtual geomagnetic pole positions of each Precambrian rock unit.

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tion, all the samples of the diabase indicate a stable component of TRM above 200 oe; i.e., the direction of magnetization and intensity remain essentially constant.

Although numerous granite exposures in the St. Francois Mountains have been mapped and are easily accessible, the collection of samples for paleomagnetic measurements presented a problem. The typical granite outcrop is very massive and usually deeply weathered, and it was often impossible to collect fresh samples. Even when fresh samples could be collected from quarries, the typical quarry granite usually did not have a component of remanent magnetization strong enough to be measured accurately with the measuring apparatus used. Therefore, no mean direction of remanent magnetization for the granites is given in Table 1.

To evaluate the effects of surface weathering and lightning on samples collected from surface exposures, we collected a suite of 10 samples from different levels of an active underground iron mine. The samples included massive hematitic ore, andesitic country rock, and vary-

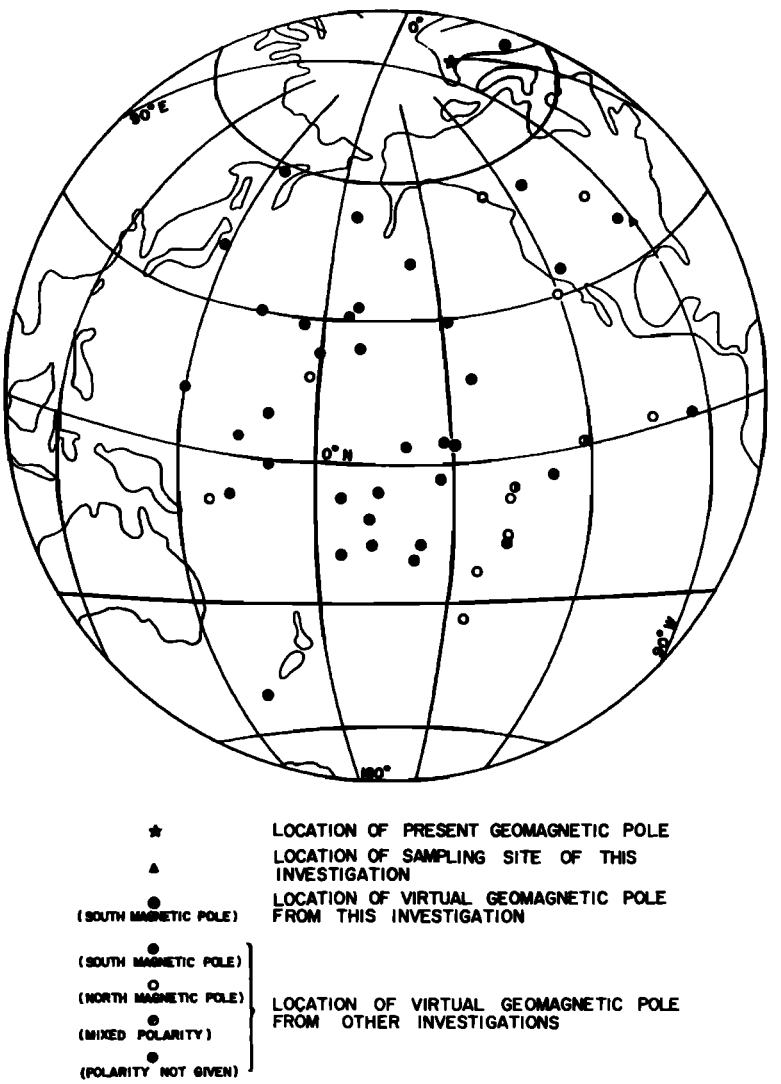


Fig. 8. Location of Precambrian virtual geomagnetic pole position of this investigation compared to results of other investigators (after *Cox and Doell* [1960]).

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ing combinations of the two. The results of the measurements are indicated in Figure 6. The remanent magnetization properties of the ore samples were particularly impressive. The ore samples displayed not only a stable component of remanent magnetization but also one which was approximately 200 times greater in intensity than the induced magnetization component. The magnitude of the remanent magnetization component undoubtedly has an important role in the aeromagnetic anomaly observed at Iron Mountain.

Virtual geomagnetic pole positions. Paleomagnetic data are generally represented in terms of the geocentric dipole which would give rise to the direction of magnetization measured in the sample. This representation provides an estimate of the departure of the orientation of the dipole axis for the past geomagnetic field (assumed also to be a geocentric dipole) from the present orientation.

The virtual geomagnetic pole positions which correspond to the mean direction of remanent magnetization of each rock unit are shown in Figure 7. The pole positions are closely grouped in the vicinity of 150°W, 5°N. Despite the limited number of Precambrian rock exposures suitable for paleomagnetic research and the relatively limited amount of research conducted to date with rocks of Precambrian age, some pertinent data have been compiled (see *Cox and Doell* [1960]). The virtual geomagnetic pole positions resulting from known investigations of the Precambrian are plotted in Figure 8, together with the mean position obtained in this investigation. Although the pole positions are scattered, the following may be concluded: First, all the Precambrian virtual geomagnetic pole positions are significantly different from the present geographical and geomagnetic pole positions. Second, the distribution of pole positions is not random; they tend to concentrate near the equator in the vicinity of 150°W. If it is valid to assume that the Precambrian geomagnetic field configuration was similar to that of the present field, a Precambrian pole would fall near the equator in the vicinity of Christmas Island.

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